

GO CLASSES
Database Management System
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Next Chapter:
The
Relational Model

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Codd proposed the relational data model in 1970. At that time most database systems were based on one of two older data models (the hierarchical model and the network model); the relational model revolutionized the database field and largely supplanted these earlier models. Prototype relational database management systems were developed in pioneering research projects at IBM and UC-Berkeley by the mid-70s, and several vendors were offering relational database products shortly thereafter. Today, the relational model is by far the dominant data model and is the foundation for the leading DBMS products, including IBM's DB2 family, Informix, Oracle, Sybase, Microsoft's Access and SQLServer, FoxBase, and Paradox. Relational database systems are ubiquitous in the marketplace and represent a multibillion dollar industry.

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The relational model is very simple and elegant; a database is a collection of one or more *relations*, where each relation is a table with rows and columns. This simple tabular representation enables even novice users to understand the contents of a database, and it permits the use of simple, high-level languages to query the data. The major advantages of the relational model over the older data models are its simple data representation and the ease with which even complex queries can be expressed.

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Relational model → Table/Relation based model

Proposed by E.F. Codd

A ₁	A ₂	A ₃

In a Research paper → Turing Award

- ① How to store the Data
- ② Terminology, Definition, Concepts
- ③ Two query languages for Relational model.

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Concepts → Concept of Normalization (2NF, 3NF, BCNF)

→ Relational Algebra

→ Relational Calculus

Two (theoretical) Query languages for Relational model

IBM ⇒ SQL

Commercial Query Language

Theoretical model

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Query languages: for retrieval of information

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A Table / Relation

Table name: Product

Column names: Name, Price, Category, Manufacturer

Name	Price	Category	Manufacturer
gizmo	\$19.99	gadgets	GizmoWorks
Power gizmo	\$29.99	gadgets	GizmoWorks
SingleTouch	\$149.99	photography	Canon
MultiTouch	\$203.99	household	Hitachi

Rows

Properties of Product table.

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A Table / Relation

Table name: Product

Column names: Name, Price, Category, Manufacturer

Name	Price	Category	Manufacturer
gizmo	\$19.99	gadgets	GizmoWorks
Power gizmo	\$29.99	gadgets	GizmoWorks
SingleTouch	\$149.99	photography	Canon
MultiTouch	\$203.99	household	Hitachi

Rows

Schema

Instance

Properties of Product table.

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Student Table:

Student → Table name
S_no, name, DOB, P.No → Column/Attribute
Field

S_no	name	DOB	P.No
1	a	20 Jan	101
2	a	20 Jan	102
3	b	20 Feb	103

Detg
Record
Tuple
Row

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Relational model → Implementation friendly

using SQL
implement

Table = Relation
Row = Record = Tuple
Column = Attribute = field

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Relational Model

- A relation: a set of records (more formally later)
- Schema: a description of data in terms of a data model
 - Name of the schema, name of each field, type of each field

SQL

```
Students(
  roll: string,
  name: string,
  age: integer,
  gpa: real
)
```

Students

roll	name	age	gpa
0001	John	23	9.1
0002	Amit	24	8.7
0003	Vivek	25	9.2
0004	Sandhya	23	9.0
...	...	...	...

Relational Model

- A relation: a set of records (more formally later)
- Schema: a description of data in terms of a data model
 - Name of the schema, name of each field, type of each field

Schema

SQL

```
Students(
  roll: string,
  name: string,
  age: integer,
  gpa: real
)
```

Students

roll	name	age	gpa
0001	John	23	9.1
0002	Amit	24	8.7
0003	Vivek	25	9.2
0004	Sandhya	23	9.0
...	...	...	...

Instance

Relations

- Relations are basically tables of data
 - Each row represents a record in the relation
- A relational database is a collection of relations
 - Each relation has a unique name
- Each row in the table specifies a relationship between the values in that row
 - The account ID "A-307", branch name "Seattle", and balance "275" are all related to each other

acct_id	branch_name	balance
A-301	New York	350
A-307	Seattle	275
A-318	Los Angeles	550
...	...	...

one
Record

account

Relations and Attributes

- Each relation has some number of attributes
 - Sometimes called "columns"
- Each attribute has a domain
 - Specifies the set of valid values for the attribute
- The *account* relation:
 - 3 attributes
 - Domain of *balance* is the set of nonnegative integers
 - Domain of *branch_name* is the set of branch names in the bank

acct_id	branch_name	balance
A-301	New York	350
A-307	Seattle	275
A-318	Los Angeles	550
...	...	...

account

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In the formal relational model terminology:

A row is called a tuple.

A column header is called an attribute.

The table is called a relation.

The data type describing the types of values that can appear in each column is represented by a domain of possible values.

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Domain of an attribute in Relational Model :

A domain D is a set of atomic values that an attribute can take.

By atomic we mean that each value in the domain is indivisible as far as the formal relational model is concerned.

Atomic $\equiv$ Indivisible

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Student Table

Domain: Strings

S-name	DOB
Ravi Kumar	20 - march

Can we Retrieve this? No
(Only surname)

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Student Table:

f-name	L-name	DOB
	Kumar	

Can be Retrieved

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Student

S-name	DOB	phono.
a	1 Jan	334 555

integer

In Relational model; we can not put ≥ 2 elements of Domain in a Row.

Atomic, indivisible element of Domain

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Student

S-name	DOB	phono.
a	1 Jan	334
a	1 Jan	555

integer

In Relational model; we can not put ≥ 2 elements of Domain in a Row.

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Schema (Structure/outline of table)

Vs

Instance (Snapshot) (Snapshot of Table)

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Schema:

- ① Table name
- ② Attributes names
- ③ Attributes Domains

Course (C-id: integer, C-name: String, faculty: String)

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The degree (or arity) of a relation is the number of attributes n of its relation schema.

Cardinality: Number of rows in any given instance of a relation.

Degree of a Relation/Table: #Columns

Cardinality r, " " " : #Records

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Tuples and Attributes

- Each row is called a tuple
 - A fixed-size, ordered list of values
- A tuple variable can refer to any valid tuple in a relation
- Each attribute in the tuple has a unique name
- Can also refer to attributes by number
 - Attribute 1 is the first attribute, etc.
- Example:
 - Tuple variable t refers to first tuple in *account* relation
 - $t[\text{balance}] = 350$
 - $t[2] = \text{"New York"}$

acct_id	branch_name	balance
A-301	New York	350
A-307	Seattle	275
A-318	Los Angeles	550
...	...	...

account

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Chapter: The Relational Model

Next Topic:

Why this model is called "relational"?

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Why call a Table, a Relation?

"Relation" (in mathematics)

binary Relation: $R: A \rightarrow B$ means $R \subseteq A \times B$

$A \times B = \{ (a, b) \mid a \in A, b \in B \}$

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Why call a Table, a Relation?

Relation (in mathematics)

ternary Relation: $R \subseteq A \times B \times C$

Ex: A : set of all countries ✓
 B : set of all players ✓
 C : { batsman; bowler; keeper } ✓

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ternary Relation: $R \subseteq A \times B \times C$

$A \times B \times C = \{ (a, b, c) \mid a \in A, b \in B, c \in C \}$

Order-Triple (3-tuple)

$A \times B \times C = \{ (SL, VK, B), (Indis, MSD, B) \dots \dots \dots (I, VK, Bat), (SL, Sanga, Bat) \dots \dots \dots \}$

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Ternary Relation: $R \subseteq A \times B \times C$

$A \times B \times C = \{ (a, b, c) \mid a \in A; b \in B; c \in C \}$

Order-Triple (3-tuple)

$R \subseteq A \times B \times C$

Ternary Relation

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$R \subseteq A \times B \times C$

Define (R): $(a, b, c) \in R$ iff b plays for a as c .

Ex: Vk " " India as Bat.
Blee " " Aus " Bowler.

$R \subseteq A \times B \times C$

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$R \subseteq A \times B \times C$

Define R: $(a, b, c) \in R$ iff b plays for a as c .

$(India, Vohra, Bat) \in R$
 $(India, Yashpal, Bat) \notin R$

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n-ary Relation:

$R \subseteq A_1 \times A_2 \times \dots \times A_n$

$A_1 \times A_2 \times \dots \times A_n = \{ (a_1, a_2, \dots, a_n) \mid a_i \in A_i \}$

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Table :

Student

Domain = String = D_1
Data = D_2
 $\{m, f\} = D_3$

S-name	DOB	gender
Chin tu	2-01-2018	m
John	10-11-2001	m

$r_1 \rightarrow$ (Chin tu, 2-01-2018, m)
 $r_2 \rightarrow$ (John, 10-11-2001, m)

$r_1 \in D_1 \times D_2 \times D_3$
 $r_2 \in D_1 \times D_2 \times D_3$

Instance

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Instance $\subseteq D_1 \times D_2 \times D_3$

Set of all Records

Table $\subseteq D_1 \times D_2 \times D_3 \times \dots \times D_n$

Relation

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Conclusion:

Table R

A_1	A_2	A_3	...	A_n
-------	-------	-------	-----	-------

Domain D_1 Domain D_2 D_3 D_n

① Every Record/Row r of R :

$r \in D_1 \times D_2 \times \dots \times D_n$

$(a_1, a_2, \dots, a_n) \in D_1 \times D_2 \times \dots \times D_n$

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Conclusion:

Table R

A_1	A_2	A_3	...	A_n
-------	-------	-------	-----	-------

Domain D_1 Domain D_2 D_3 D_n

② Every Instance I of R :

set of records

$I \subseteq D_1 \times D_2 \times \dots \times D_n$

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Conclusion:

Table R

Relation

$R \subseteq D_1 \times D_2 \times \dots \times D_n$

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The Relational Data Model (E.F. Codd – 1970)

- The Relational Data Model uses the mathematical concept of a **relation** as the formalism for describing and representing data.
- Question:** What is a relation?
- Answer:**
 - Formally, a **relation** is a subset of a cartesian product of sets.
 - Informally, a relation is a **"table"** with rows and columns.

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The Relational Data Model (E.F. Codd – 1970)

- The Relational Data Model uses the mathematical concept of a **relation** as the formalism for describing and representing data.
- Question:** What is a relation?
- Answer:**
 - Formally, a **relation** is a subset of a cartesian product of sets.
 - Informally, a relation is a **"table"** with rows and columns.

CHECKING-ACCOUNT Table

branch-name	account-no	customer-name	balance
Orsay	10991-06284	Abiteboul	\$3,567.53
Hawthorne	10992-35671	Hull	\$11,245.75
...	...	...	...

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- Most database systems use the relational model
 - Record-based model
 - Collection of tables containing records
 - Format of records is fixed
 - It can be changed, but this is infrequent!
 - Data is modeled at logical level, not physical level

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Review of Mathematical Concepts (1)

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- **Sets:** $S, T, S_1, \dots, S_n, T_1, \dots, T_n, \{\}$
Cardinality of a set S denoted as $|S|$
- **Cartesian Product** of sets (also *cross product*):
 $S \times T$ set of all pairs (s, t) where $s \in S$ and $t \in T$
 $S_1 \times \dots \times S_n$ set of all n -tuples $(s_1, \dots, s_n)$
where $s_i \in S_i$
- **Relation R** over S, T :
subset of $S \times T$, written $R \subseteq S \times T$
We write $(s, t) \in R$ or, equivalently, sRt

Review of Mathematical Concepts (2)

- **Relation R** over $S_1, \dots, S_n$:
subset $R \subseteq S_1 \times \dots \times S_n$
The number n is the **arity** of R
(R is **binary** if $n=2$ and **ternary** if $n=3$)
- **Function f** from S to T , written $f: S \rightarrow T$
associates to every element $s \in S$
exactly one element of T , denoted as $f(s)$

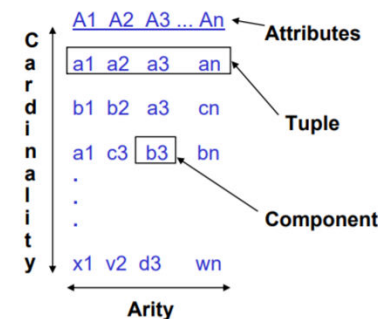
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Basic Notions from Discrete Mathematics

- A **k -tuple** is an ordered sequence of k objects (need not be distinct)
□ $(2, 0, 1)$ is a 3-tuple; (a, b, a, a, c) is a 5-tuple, and so on.
- If $D_1, D_2, \dots, D_k$ are k sets, then the **cartesian product** $D_1 \times D_2 \times \dots \times D_k$ of these sets is the set of all k -tuples $(d_1, d_2, \dots, d_k)$ such that $d_i \in D_i$, for $1 \leq i \leq k$.
- **Fact:** Let $|D|$ denote the cardinality (= # of elements) of a set D .
Then $|D_1 \times D_2 \times \dots \times D_k| = |D_1| \times |D_2| \times \dots \times |D_k|$.
- **Example:** If $D_1 = \{0, 1\}$ and $D_2 = \{a, b, c, d\}$, then $|D_1| \times |D_2| = 8$.

Tables Look Like Relations ...

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$\{(a_1, a_2, a_3, \dots, a_n),$
 $(b_1, b_2, a_3, \dots, c_n),$
 $(a_1, c_3, b_3, \dots, b_n),$
 $\dots$
 $(x_1, v_2, d_3, \dots, w_n)\}$

Over which sets does this relation range?

- In Databases:** Distinguish between
- **Schema** (structure) and
 - **Instance** (content)

Relation Schemas

A **relation schema**

$R(A_1:D_1, \dots, A_n:D_n)$

consists of

- a **name**, say R
- a nonempty set of **attributes**, say $A_1, \dots, A_n$
- a **type** or **domain**, say $D_i = \text{dom}(A_i)$, for each attribute A_i

Example: Product (Prodname: Name,
Price: DollarPrice,
Category: Name,
Manufacturer: Name)

Φ : Relation R Schema:

Table $R(A_1:D_1, A_2:D_2, A_3:D_3)$ (Domain of A_2)

Assume: $|D_1| = 2$
 $|D_2| = 3$
 $|D_3| = 3$

① Cardinality of Largest Instance?

$\rightarrow$ #Records = 18

largest instance $I = D_1 \times D_2 \times D_3$

$2 \times 3 \times 3 = 18$

$\Leftarrow$ #elements = $|D_1| \times |D_2| \times |D_3|$

② #instances possible?

2^{18}

To create Instance:

for every record in $D_1 \times D_2 \times D_3$
we have 2 choices: Take / Don't Take

Types and Domains

Type: Class of atomic values, e.g.,

- integers, reals, strings
- integers between 15 and 80, strings of (up to) 50 characters

Domain: Set of atomic values, that have a specific meaning in an application, e.g.,

– Name, EmployeeAge

- Domains have a **type**, e.g.,
– EmployeeAge = Int[15,80]

Example

Domain declaration:

Name=String(30), DollarPrice=Decimal (10,2),

Relation schema:

Product(Prodname: Name, Price: DollarPrice,
Category: Name, Manufacturer: Name)

Instance:

Prodname	Price	Category	Manufacturer
gizmo	19.99	gadgets	GizmoWorks
Power gizmo	29.99	gadgets	GizmoWorks
SingleTouch	149.99	photography	Canon
MultiTouch	203.99	household	Hitachi

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Empty Relation = Empty Instance
= No Records

Null Values

Attribute values

- are atomic
- have a known domain
- can sometimes be “null”

Three meanings of null values

1. not applicable
2. not known
3. absent (not recorded)

Student

studno	name	hons	tutor	year	thesis title
s1	jones	ca	bush	2	null
s2	brown	cis	kahn	2	null
s3	smith	null	goble	2	null
s4	bloggs	ca	goble	1	null
s5	jones	cs	zobel	1	null
s6	peters	ca	kahn	3	“A CS Survey”

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School: → Student Schema: outline

S-name	Passport no	Gender
a	101112	M
b	NULL	F

→ means for b; PNo info is NOT Available

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College:

NOT applicable

NOT available

Shame	gender	Husband name	wife name
John	M	NULL	NULL
Deepika	F	Ranveer	NULL

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In Relational model:

Table $\equiv$ Relation

→ Set

Order of Rows does not matter

↓

Set of Records

↓

Unordered; No Duplicates

↓

No Two Rows can be Entirely same.

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In mathematical relations:

- Order of tuples and duplicate tuples do not matter
- Order of attributes is still fixed

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Relation $\subseteq D_1 \times D_2 \times \dots \times D_n$

every record $\rightarrow$ ordered n-tuple

So Column Order matters.

will see it in Relational Algebra.

Basic Notions from Discrete Mathematics

- A **k-ary relation** R is a subset of a cartesian product of k sets, i.e.,
 - $R \subseteq D_1 \times D_2 \times \dots \times D_k$.
- Examples:
 - Unary $R = \{0, 2, 4, \dots, 100\}$ ($R \subseteq D$)
 - Binary $T = \{(a, b) : a \text{ and } b \text{ have the same birthday}\}$
 - Ternary $S = \{(m, n, s) : s = m + n\}$
 - ...

Relations and Attributes

- Note:**
 $R \subseteq D_1 \times D_2 \times \dots \times D_k$ can be viewed as a table with k columns

Table S

3	5	8
150	100	250
...	...	...

- In the relational data model, we want to have names for the columns; these are the **attributes** of the relation.

Tuples and Relationships

- In the *account* relation:
 - Domain of *acct_id* is D_1
 - Domain of *branch_name* is D_2
 - Domain of *balance* is D_3
- The *account* relation is a subset of the tuples in the Cartesian product $D_1 \times D_2 \times D_3$
- Each tuple included in *account* specifies a relationship between that set of values
 - Hence the name, "relational model"
 - Tuples in *account* relation specify the details of valid bank accounts

acct_id	branch_name	balance
A-301	New York	350
A-307	Seattle	275
A-318	Los Angeles	550
...	...	...

account

Tuples and Relations

- A relation is a set of tuples
 - Each tuple appears exactly once
 - Note: SQL tables are multisets!
 - If two tuples t_1 and t_2 have the same values for all attributes, then t_1 and t_2 are the *same tuple* (i.e. $t_1 = t_2$)
- The order of tuples in a relation is not relevant

Relation Schemas

- Every relation has a schema
 - Specifies the type information for relations
 - Multiple relations can have the same schema
- A relation schema includes:
 - a set of attributes
 - the domain of each attribute
- Naming:
 - Relation names written as all lowercase
 - Relation schema's name is capitalized
- For relation r and relation schema R :
 - Write $r(R)$ to indicate that schema of r is R

Relational Database Schemas - Examples

- BANKING relational database schema with relation schemas
 - CHECKING-ACCOUNT(branch, acc-no, cust-id, balance)
 - SAVINGS-ACCOUNT(branch, acc-no, cust-id, balance)
 - CUSTOMER(cust-id, name, address, phone, email)
 -
- UNIVERSITY relational database schema with relation schemas
 - STUDENT(student-id, student-name, major, status)
 - FACULTY(faculty-id, faculty-name, dpt, title, salary)
 - COURSE(course-no, course-name, term, instructor)
 - ENROLLS(student-id, course-no, term)
 - ...

Schema of *account* Relation

- The relation schema of *account* is:
 $Account_schema = (acct_id, branch_name, balance)$

- To indicate that *account* has schema $Account_schema$:
 $account(Account_schema)$

acct_id	branch_name	balance
A-301	New York	350
A-307	Seattle	275
A-318	Los Angeles	550
...	...	...

account

- Note:
 - Domains aren't typically stated explicitly in this notation...

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Ex: School:

Student

S-name	Pass No

Not Null

maybe Null

Implement using SQL: CREATE TABLE Student
{ S-name string NOT Null
Passno Integer;

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Chapter: The Relational Model

Next Topic:

Keys

Candidate Keys

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Storing Data:

Sname	DOB	A	B
a	1 mar		
b	1 feb	1	2
a	1 feb	1	1
a	1 Jan	2	1
		3	4
		3	5

AB
2 1

(AB)

Different value in Every Tuple

10K
Records

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Super key:

can be singleton also

a set of attributes which can identify any record uniquely.

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Super key : can be singleton also
a set of attributes which
different values in different
records.

No Two Records can have same value
for S.

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Table :

Sname	State	Dist	pincode	HNo
A	TN		101	30
A	Rajasthan		102	30

Superkey

Super key

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Table :

A B C D E

A B Super key

C Super key

D E Super key

X {

{

a₁ b₁

a₁ b₂

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Table :

Sname Aadhar-no Pan-No pincode HNo

Superkey

Super key

Superkey

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Q: In Relational model;
at least one super key is
Definitely Present?

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Q: In Relational model;
at least one super key
Definitely Present? → True

In worst case; all attributes combined
will work as super key.

Two records can never be entirely same

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In Relational model;
Set of all attributes is
Always a Superkey.

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In Relational model:

Table = Relation → a "set" of records

So Two records can never be same.

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Table:

A	B	C
1	1	2
1	1	2
1	1	3

Handwritten notes: A red arrow points from the header 'A B C' to a small table below it:

A	B	C
1	1	3

A purple bracket groups the first two rows of the main table, and a purple arrow points from this bracket to the value '3' in the third row, column 'C'.

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Table:

A	B	C	D
1	1	1	
2	1	1	
3	1	2	
4	2	3	

Handwritten notes: 'A is a Unique (superkey)' with an arrow pointing to the 'A' column. 'AB = Superkey' with an arrow pointing to the 'A' and 'B' columns. 'Redundant' with an arrow pointing to the 'C' column. 'AB Superkey' with an arrow pointing to the 'A' and 'B' columns.

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Candidate key (ck):

$\Rightarrow$ a minimal super key

$\Rightarrow$ If $CK = A_1, A_2, \dots, A_n$ then $\forall_i (CK - A_i)$ is not a superkey.

Handwritten notes: 'minimal super key' with an arrow pointing to the 'minimal' in the definition. 'then' with an arrow pointing to the 'then' in the definition.

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Student:

name	adhar-no	pan-no.

Handwritten notes: 'superkey (not minimal super key)' with an arrow pointing to the 'adhar-no' and 'pan-no.' columns. 'adhar-no itself is superkey' with an arrow pointing to the 'adhar-no' column. 'Cand-key' with an arrow pointing to the 'name' column. 'Cand key' with an arrow pointing to the 'adhar-no' column.

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Super key :

Super key must identify all the "non-null" records uniquely.

→ Null values are Allowed.

Super key :

Super key must have Different values for the "non-null" records in two tuples.

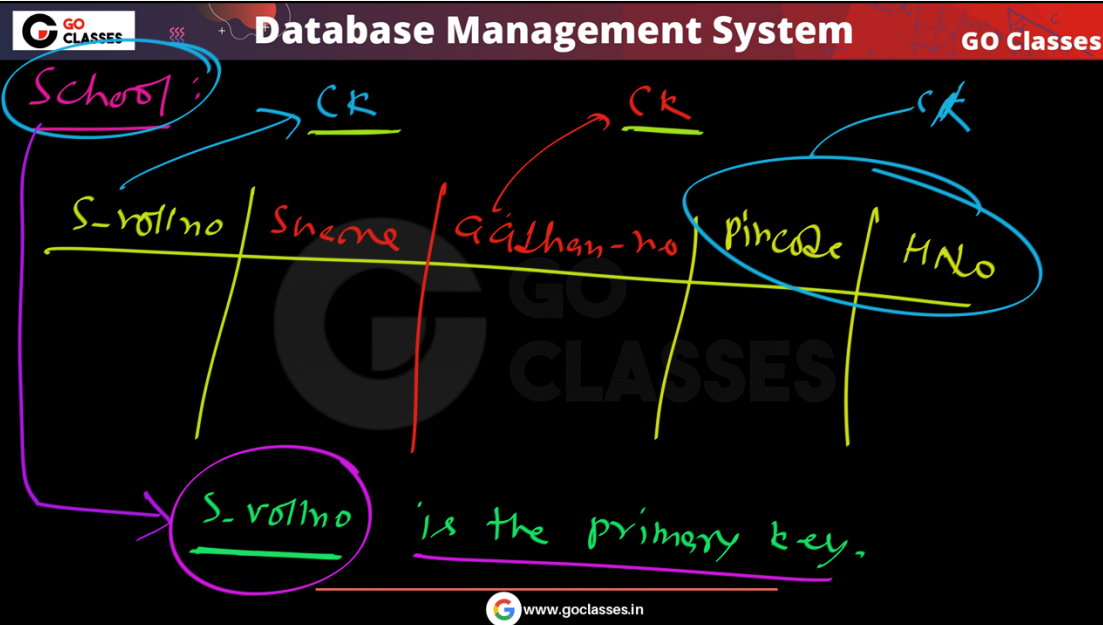
→ Null values are Allowed.

Candidate key :

Cand. key must have Different values for the "non-null" records in two tuples.

and must be minimal.

→ Null values are Allowed.



Primary key:

One of the Candidate key is Chosen as (selected as) to be a primary key which must be NOT NULL.

School:

S-rollno	Sname	Passport no	Pincode	HNo
a		1011		
b		NULL		
c		2022		
d		NULL		

Annotations: CK (Candidate Key) points to S-rollno, Sname, Passport no, Pincode, and HNo. A note states: "Can NOT be chosen as PK" with an arrow pointing to the Passport no column. Another note states: "unique" with an arrow pointing to the Passport no column.

Note:

If PK is multiple attributes then No attribute of PK can have NULL values.

Note:

In relational model Any Relation, at any point of time, must have Exactly one PK.

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Candidate key : most Important

Minimal Super key

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Candidate key :

- Simple CK : Simple attribute CK
- Composite CK : ≥ 2 attributes CK

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Candidate keys

CK1 CK2 ... CKn

Primary key

Alternate keys

Can Never be Null

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Distinguishing Tuples

- Relations are sets of tuples...
 - No two tuples can have the same values for all attributes...
 - But, some tuples might have the same values for some attributes
- Example:
 - Some accounts have the same balance
 - Some accounts are at the same branch

acct_id	branch_name	balance
A-301	New York	350
A-307	Seattle	275
A-318	Los Angeles	550
A-319	New York	80
A-322	Los Angeles	275

account

Keys

- Keys are used to distinguish individual tuples
 - A superkey is a set of attributes that uniquely identifies tuples in a relation

- Example:

{ *acct_id* } is a superkey

acct_id	branch_name	balance
A-301	New York	350
A-307	Seattle	275
A-318	Los Angeles	550
A-319	New York	80
A-322	Los Angeles	275

account

- Is { *acct_id*, *balance* } a superkey?
 - Yes! Every tuple has a unique set of values for this combination of attributes.
- Is { *branch_name* } a superkey?
 - No. Each branch can have multiple accounts

Superkeys and Candidate Keys

- A superkey is a set of attributes that uniquely identifies tuples in a relation
- Adding attributes to a superkey produces another superkey
 - If { *acct_id* } is a superkey, so is { *acct_id*, *balance* }
 - If a set of attributes $K \subseteq R$ is a superkey, so is any superset of K
 - Not all superkeys are equally useful...
- A *minimal* superkey is called a candidate key
 - A superkey for which no proper subset is a superkey
 - For *account*, only { *acct_id* } is a candidate key

Primary Keys

- A relation might have several candidate keys
- In these cases, one candidate key is chosen as the primary means of uniquely identifying tuples
 - Called a primary key

- Example: *customer* relation

- Candidate keys could be:

{ *cust_id* }

{ *cust_name*, *address* }

- Choose primary key:

{ *cust_id* }

cust_id	cust_name	address
23-652	Joe Smith	123 Evergreen Way
15-202	Ellen Jones	35 Market St. #214
23-521	Dave Johnson	621 Washington Rd.
...	...	...

customer

The primary key attributes are required to be nonnull and unique; that is, no tuple can have a null value for a primary-key attribute, and no two tuples in the relation can be equal on all the primary-key attributes.

Primary Keys (2)

- Keys are a property of the relation schema, not individual tuples
 - Applies to *all* tuples in the relation
- Primary key attributes are listed first in relation schema
 - Key attributes are also underlined
- Examples:
 - $Account_schema = (\underline{acct_id}, branch_name, balance)$
 - $Customer_schema = (\underline{cust_id}, cust_name, address)$
- Only indicate primary keys in this notation
 - Other candidate keys are not specified

Primary Keys (3)

- Multiple records can't have the same values for a primary key!
- Example:

cust_id	cust_name	address
23-652	Joe Smith	123 Evergreen Way
15-202	Ellen Jones	35 Market St. #214
23-521	Dave Johnson	621 Washington Rd.
15-202	Albert Stevens	92 Terrace Lane
...	...	...

✗

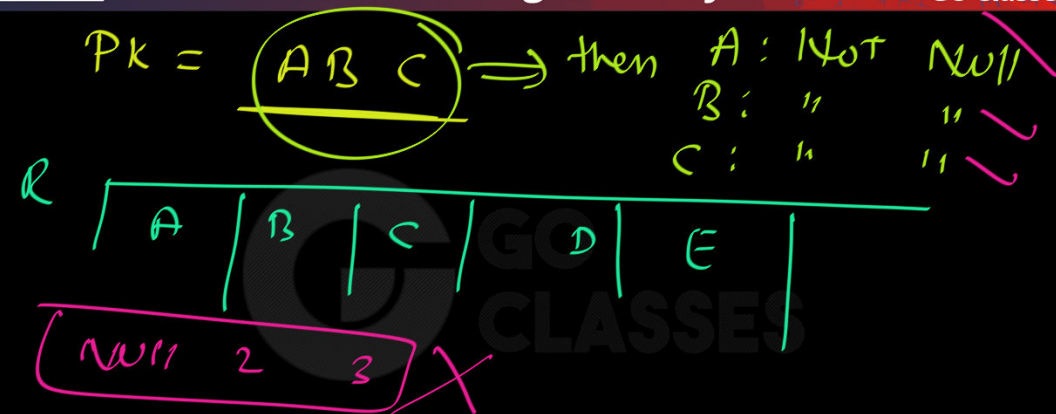
customer

- Two customers cannot have the same ID.

Keys Constrain Relations

- Primary keys *constrain* the set of tuples that can appear in a relation
 - Same is true for *all* superkeys
- For a relation r with schema R
 - If $K \subseteq R$ is a superkey then
 - $\langle \forall t_1, t_2 \in R : t_1[K] = t_2[K] : t_1[R] = t_2[R] \rangle$
 - i.e. if two tuples have the same values for the superkey attrs, then they are the same tuple

Database Management System



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Q: In Rel. model;
Can a Complete tuple be Null?

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Q: In Rel. model; $\rightarrow$ Exactly one
Primary key
Can a Complete tuple be Null?

$\exists$ (pk) $\rightarrow$ NOT Null

$\rightarrow$ No (NEVER)

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Rel. model
ER model
SQL

Different

$\rightarrow$ have their Differences

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Note:

Ck must Identify
"non-null" tuples uniquely.

Null values Allowed in Ck.

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Note: PK is one of the CKs which must be NOT Null.

→ Any such CK can be chosen as PK (DB owner decides)

Keys: Definitions

- **Superkey**
 - a set of attributes whose values together *uniquely* identify a tuple in a relation
- **Candidate Key**
 - a superkey for which no proper subset is a superkey: a superkey that is *minimal*
 - Can be more than one for a relation
- **Primary Key**
 - a candidate key chosen to be the main key
 - One for each relation, indicated by underlining the key attributes

Student(studno, name, tutor, year)

Example: Multiple Keys

Student (Lastname, Firstname, MatriculationNo, Major)

⏟
Candidate key
(2 attributes)

⏟
Candidate key
⏟
Superkey

Note: There are alternate candidate keys

- Candidate keys are
{Lastname, Firstname} and
{StudentID}